Generalized linear models

Models for binary outcome

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Models for binary outcome - Contents

Models for binary outcome

Binary outcome

Let $y \in \{0, 1\}$ be a binary response variable. Is

$$\mathbb{E}(y \mid \mathbf{x}) = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta}$$

reasonable?

Binary outcome

Reasons that speak against this model approach:

Motivating the binary model via Bayes' theorem

Let $(y, \mathbf{x})^\top$ be a random vector with

$$y \in \{0, 1\}, \qquad \mathbf{x} = (x_1, \ldots, x_p)^\top$$

and $\mathbf{x} \mid y = i \sim \mathrm{N}(\boldsymbol{\mu_i}, \boldsymbol{\Sigma})$.

Recall: Bayes' theorem

$$P(y \mid x) = \frac{P(x \mid y)P(y)}{P(x)}$$

Motivating the binary model via Bayes' theorem

For continuous x and binary y:

$$P(y = 1 \mid x) = \frac{f(x|y{=}1)P(y{=}1)}{f(x)} =$$

Motivating the binary model via Bayes' theorem

Taking the logarithm of the numerator provides

Motivating the binary model via Bayes' theorem

Plugging this into the fraction

Motivating the binary model via latent variables

Basic idea:

- We observe a binary response $y \in \{0, 1\}$
- However, we assume underlying continuous variable

$\rightarrow$ Not observable or unknown: latent

- Actually, continuous $\tilde{y}$ in the background

Motivating the binary model via latent variables

Example

Let

- y_i: response variable
- x_i: covariate
- $\tilde{y}_i$: latent metric variable

For example:

- y_i:
- x_i:
- $\tilde{y}_i$:

Or:

- y_i:
- x_i:
- $\tilde{y}_i$:

Motivating the binary model via latent variables

Assumption

$\tilde{y}_i$ follows the classical linear model

$$\tilde{y}_i = \gamma_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i \,,$$

where $-\varepsilon_i$ has the distribution function F.

Motivating the binary model via latent variables

Relationship between y and $\tilde{y}$

Observation		and		latent variable
$y_i = 1$		$\Leftrightarrow$		$\tilde{y}_i \geq \vartheta$

For example:

$\Leftrightarrow$

Motivating the binary model via latent variables

Then we obtain:

$P(y_i = 1 \mid x_i) =$

Motivating the binary model via latent variables

Let

$$\pi_i(\mathbf{x}_i) = P(y_i = 1 \mid \mathbf{x}_i) = F(\beta_0 + \mathbf{x}_i^\top \beta)$$

Depending on the choice of $F(\cdot)$ different models result:

1. $F(\cdot) =$
2. $F(\cdot) =$
3. $F(\cdot) =$